

3) Unit Vector Conversion

→ same as above

Dot Product in diff. C.S

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

	$\hat{a}_x$	$\hat{a}_y$	$\hat{a}_z$
$\hat{a}_x$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\hat{a}_y$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\hat{a}_z$	$\cos \theta$	$-\sin \theta$	0

LOCUS (How)

VECTOR CALCULUS

- # Differential Length $d\vec{l} \Rightarrow$ vector parameter
- # Differential Surface $d\vec{S} \Rightarrow$ vector parameter
- # Differential Volume $dV \Rightarrow$ scalar parameter

* RECTANGULAR COORDINATE Sys.

- $$d\vec{l} = dx \hat{a}_x + dy \hat{a}_y + dz \hat{a}_z$$

$\hookrightarrow$ differential length

- differential Surface :

$$x = \text{const.} \Rightarrow$$

$$d\vec{S} = dy dz (\pm \hat{a}_x)$$

$$y = \text{const.} \Rightarrow$$

$$d\vec{S} = dx dz (\pm \hat{a}_y)$$

$$z = \text{const.} \Rightarrow$$

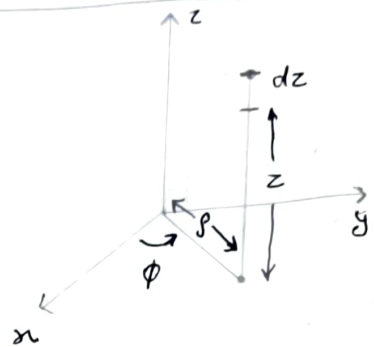
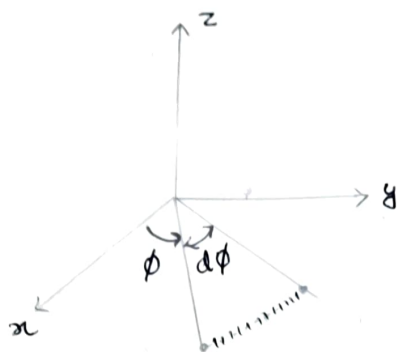
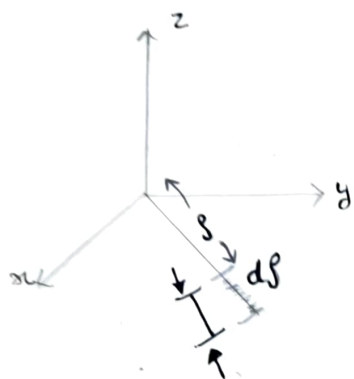
$$d\vec{S} = dy dx (\pm \hat{a}_z)$$

- differential Volume :

$$dV = dx dy dz$$

CYLINDRICAL COORDINATE SYSTEM

Differential length : $\vec{dl} = dr \hat{a}_r + r d\phi \hat{a}_\phi + dz \hat{a}_z$



Differential surface : $d\vec{u} = r dr \hat{a}_r + r d\phi \hat{a}_\phi + dz \hat{a}_z$

i) $r = \text{const. surface} \Rightarrow d\vec{S} = r d\phi dz (\hat{a}_r)$

ii) $\phi = \text{const. surface} \Rightarrow d\vec{S} = dr dz (\pm \hat{a}_\phi)$

iii) $z = \text{const. surface} \Rightarrow d\vec{S} = r dr d\phi (\pm \hat{a}_z)$

Differential Volume :

$dV = r dr d\phi dz$

SPHERICAL COORDINATE SYSTEM

$\vec{dl} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$ ^{**} $\{ r = r \sin\theta \}$

$\Rightarrow d\vec{u} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$

Differential Surface : $d\vec{l} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$

- i) $r = \text{const. surface} \Rightarrow d\vec{S} = r^2 \sin\theta d\theta d\phi (\hat{a}_r)$
 ii) $\theta = \text{const. surface} \Rightarrow d\vec{S} = r \sin\theta dr d\phi (\pm \hat{a}_\theta)$
 iii) $\phi = \text{const. surface} \Rightarrow d\vec{S} = r dr d\theta (\pm \hat{a}_\phi)$

Differential Volume

$$d\vec{l} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$$

$$dV = dr d\theta d\phi (r^2 \sin\theta)$$

NOTE :- # always consider $d\vec{l} = dr \hat{a}_r + dy \hat{a}_y + dz \hat{a}_z$

$$d\vec{l} = dy \hat{a}_y + y d\phi \hat{a}_\phi + dz \hat{a}_z$$

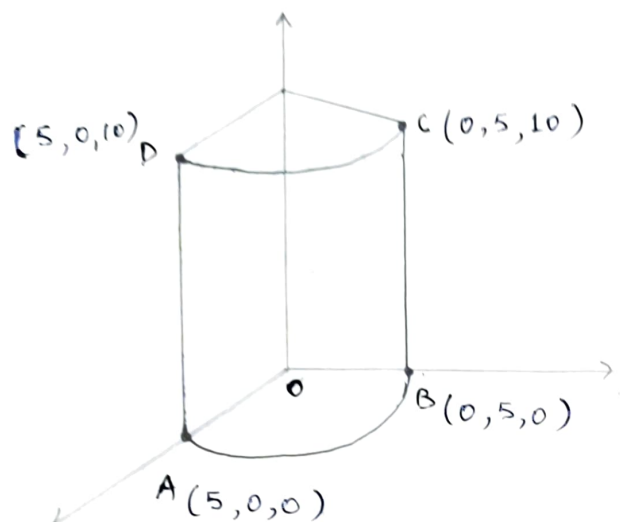
$$d\vec{l} = r dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$$

always use diff. lengths +ve, use the limits accordingly.

Ex:- $\vec{BC} = \vec{L} = \int_B^C d\vec{l}$, $\vec{CB} = \vec{L} = \int_C^B d\vec{l}$

↓
 $d\vec{l}$ should be same

Problem Solving Technique



Cylindrical Coordinates

$$A(5, 0, 0) \rightarrow A(5, 0, 0)$$

$$B(0, 5, 0) \rightarrow B(5, \frac{\pi}{2}, 0)$$

$$C(0, 5, 10) \rightarrow C(5, \frac{\pi}{2}, 10)$$

$$D(5, 0, 10) \rightarrow D(5, 0, 10)$$

$$B(5, \frac{\pi}{2}, 0) \rightarrow C(5, \frac{\pi}{2}, 10)$$

$$\rho \rightarrow 5 \text{ to } 5 \Rightarrow d\rho = 0 \Rightarrow \rho = 5$$

$$\phi \rightarrow \frac{\pi}{2} \text{ to } \frac{\pi}{2} \Rightarrow d\phi = 0 \Rightarrow \phi = \frac{\pi}{2}$$

$$z \rightarrow 0 \text{ to } 10 \Rightarrow dz \neq 0 \Rightarrow dz$$

i) For $\vec{BC}$ ($B \rightarrow C$)

$$d\vec{u} = dz \hat{a}_z$$

$$\vec{L} = \vec{BC} = \int_B d\vec{u}$$

$$= \int_{z=0}^{10} dz \hat{a}_z = [z]_0^{10} \hat{a}_z \Rightarrow \boxed{\vec{BC} = 10 \hat{a}_z}$$

$$\{|\vec{BC}| = 10\}$$

ii) For $\vec{CD}$ ($C \rightarrow D$)

$$C(5, \frac{\pi}{2}, 10) \rightarrow D(5, 0, 10)$$

$$\rho \rightarrow 5 \text{ to } 5 \Rightarrow d\rho = 0$$

$$\phi \rightarrow \frac{\pi}{2} \text{ to } 0 \Rightarrow d\phi \neq 0$$

$$z \rightarrow 10 \text{ to } 10 \Rightarrow dz = 0$$

$$d\vec{u} = \cancel{\rho d\rho \hat{a}_\rho} + \rho d\phi \hat{a}_\phi + \cancel{dz \hat{a}_z}$$

$$d\vec{u} = \rho d\phi \hat{a}_\phi$$

$$\vec{CD} = \int_C^D d\vec{u} = \int_{\pi/2}^0 \rho d\phi \hat{a}_\phi = \rho [\phi]_{\pi/2}^0 \hat{a}_\phi$$

$$\boxed{\vec{CD} = -\frac{5\pi}{2} \hat{a}_\phi}$$

$$\{|\vec{CD}| = \frac{5\pi}{2}\}$$

Note :- Calculation of $\vec{dS}$

$$\vec{dS} = \text{magnitude (dir}^n)$$

Π

I (+ve Π -ve selection)

$$\vec{S} = \int \int \vec{dS} = \int_{\text{lower}}^{\text{higher}} \oint \vec{dS}$$

3) Area of surface ABCD : # $\rho = \text{constant} = 5$

$$d\vec{l} = ds \hat{a}_s + s d\phi \hat{a}_\phi + dz \hat{a}_z$$

$$\therefore d\vec{S} = s d\phi dz (\hat{a}_s)$$

$$\vec{S} = \int \int d\vec{S} = \int \int s d\phi dz \hat{a}_s$$

$$= 5 \int_0^{\pi/2} d\phi \int_0^{10} dz \hat{a}_s$$

$$\{|\vec{S}| = 25\pi\}$$

$$\vec{S} = 5 \times \frac{\pi}{2} \times 10 = 25\pi \hat{a}_s \text{ sq. unit}$$

4) Surface ABO : # $z = \text{constant} = 0$

$$d\vec{S} = s ds d\phi (\pm \hat{a}_z) = s ds d\phi (-\hat{a}_z)$$

$$\vec{S} = -\hat{a}_z \int_0^5 s ds \int_0^{\pi/2} d\phi = -\hat{a}_z \left[\frac{s^2}{2} \right]_0^5 \left[\phi \right]_0^{\pi/2}$$

$$= -\hat{a}_z \left[\frac{25}{2} \right] \left[\frac{\pi}{2} \right] = \frac{25\pi}{4} (-\hat{a}_z)$$

Ques: A part of sphere is mentioned below:

$$3 \leq r \leq 5$$

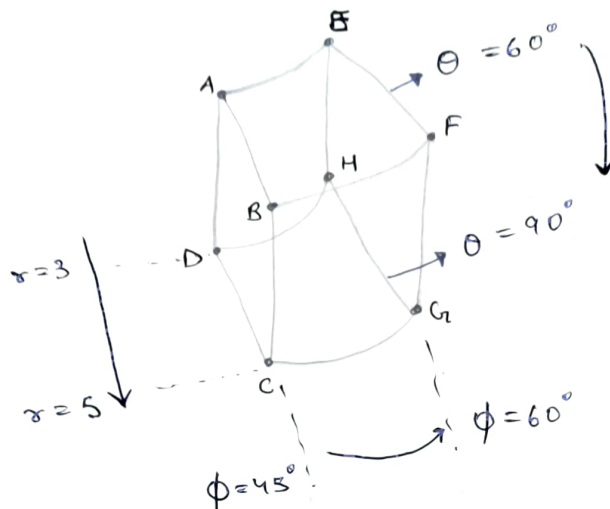
$$60^\circ \leq \theta \leq 90^\circ$$

$$45^\circ \leq \phi \leq 60^\circ$$

$$r=3 \text{ is AEHD}$$

$$\phi = 45^\circ \text{ is ABED}$$

$$\theta = 60^\circ \text{ is AEFB}$$



i) length $\vec{DH}$, $d\vec{l} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$

along $\vec{DH}$: $r=3 \rightarrow dr=0$

$\theta = 90^\circ \rightarrow d\theta=0$

$\phi = 45^\circ \text{ to } 60^\circ$

$\therefore d\vec{l} = r \sin\theta d\phi \hat{a}_\phi$

$d\vec{l} = r \sin\theta d\phi \hat{a}_\phi$

$$\vec{L} = \vec{DH} = \int_D^H d\vec{l} = \int_{45^\circ}^{60^\circ} r \sin\theta d\phi \hat{a}_\phi$$

$$= 3 \sin 90^\circ (60^\circ - 45^\circ) \hat{a}_\phi = 3 \times (15^\circ) \hat{a}_\phi$$

$$|\vec{DH}| = \frac{3 \times 15 \times \pi}{180^\circ}$$

ii) length $\vec{FG}$, $d\vec{l} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$

$r=5 \rightarrow dr=0$

$\phi = 60^\circ \rightarrow d\phi=0$

$\theta = 60^\circ \text{ to } 90^\circ$

$d\vec{l} = r d\theta \hat{a}_\theta$

$$\vec{L} = \int_{60}^{90} r d\theta \hat{a}_\theta = 5 (30^\circ) = 5 \left(\frac{\pi}{6} \right)$$

$$\vec{L} = \frac{5\pi}{6} \hat{a}_\theta$$

iii) Surface Area AEMD :

$$r = 9 \Rightarrow r = \text{const.} \quad d\vec{l} = dr\hat{a}_r + r d\theta\hat{a}_\theta + r d\phi\hat{a}_\phi$$

$$\begin{aligned} d\vec{s} &= r^2 \sin\theta d\theta d\phi (\pm \hat{a}_r) \left\{ \begin{array}{l} \text{-ve becuz dir}^n \\ \text{must be outside} \\ \text{from volume} \end{array} \right\} \\ &= r^2 \sin\theta d\theta d\phi (-\hat{a}_r) \end{aligned}$$

$$\vec{S} = r^2 \int_{\theta=60^\circ}^{90^\circ} \sin\theta d\theta \int_{\phi=45^\circ}^{60^\circ} d\phi (-\hat{a}_r)$$

$$= 9 \left[-\cos\theta \right]_{60^\circ}^{90^\circ} \left[\phi \right]_{45^\circ}^{60^\circ} = 9 \left[-\cos 90^\circ + \cos 60^\circ \right] \left[15^\circ \right]$$

$$= 9 \left[0 + \frac{1}{2} \right] \left[\frac{15 \times \pi}{180} \right] = \boxed{-\frac{3\pi}{8} \hat{a}_r}$$

iv) Volume :

$$V = \iiint dV = \iiint r^2 \sin\theta dr d\theta d\phi$$

$$V = \int_{r=3}^{r=5} r^2 dr \int_{\theta=60^\circ}^{\theta=90^\circ} \sin\theta d\theta \int_{\phi=45^\circ}^{\phi=60^\circ} d\phi =$$